TaxPadi

CODEQUEST PROJECT PROPOSAL

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1. UNDERSTANDING THE PROBLEM

Ghana's self-employed individuals, freelancers, traders, and small business owners form the backbone of the national economy. Yet the overwhelming majority operate without any system for tracking their income, understanding their tax obligations, or preparing for GRA deadlines. The result is a recurring annual crisis: last-minute scrambling, incorrect filings, missed deadlines, and compounding penalties that are entirely avoidable.

The GRA Taxpayers App addresses return filing and payment but solves the wrong problem. It assumes the user already knows their numbers. Nobody has built the tool that helps users build and understand those numbers throughout the year. That is the gap TaxPadi fills.

The Four Core Pain Points

- No financial visibility: users cannot tell what they owe GRA at any point during the year. Tax liability only becomes real when deadlines arrive, by which point it is too late to prepare accurately.
- No expense tracking: legitimate deductible expenses go unrecorded. Users consistently overpay taxes on income they could have legally reduced through proper expense claims.
- Tax deadline panic: Ghana's tax calendar spans multiple tax types with different deadlines. Most individuals are unaware of all of them and arrive at each one completely unprepared.
- Payment friction: to pay taxes, a user must first file a return on the GRA Taxpayers Portal which generates a tax bill, then navigate to ghana.gov to initiate payment, then choose between Mobile Money, bank card, USSD (*222#), a physical bank branch, or a GRA Taxpayer Service Centre in person. New taxpayers must additionally visit a GRA office to register their tax type before they can even begin filing. This multi-platform, multi-step process is confusing and a meaningful barrier to compliance, especially for users who are not digitally confident.

Why Existing Solutions Fail

- Manual spreadsheets: error prone, time consuming, and requires financial literacy most users do not have.
- Paper records: easily lost, impossible to calculate from accurately, and useless in a GRA audit.
- Hiring an accountant: expensive and inaccessible to small informal operators.
- GRA Taxpayers App: files returns but does not calculate, track, or prepare any of the underlying numbers.

The Adoption Challenge

A significant segment of the target market has limited smartphone literacy. TaxPadi addresses this through four channels: WhatsApp integration for users who prefer not to download a standalone app; USSD support for feature phone users without internet access; a commissioned agent network deployed at markets and GRA offices; and voice input in Twi, Ga, Hausa, and Ewe to eliminate the typing barrier entirely.

2. THE SOLUTION

TaxPadi is a mobile-first personal tax management platform built for Ghana's self-employed individuals, freelancers, traders, and small business owners. It gives users a complete, intelligent system for tracking their finances, understanding their tax liability, and paying what they owe - directly from their phone.

> *TaxPadi is not a filing portal. It is the financial infrastructure that makes correct filing inevitable.*

The Core Product Loop: Track - Know - Save - File - Pay

- Track: user logs income and expenses via manual entry, voice input, receipt scanning, or MoMo statement import.
- Know: a live tax liability meter on the dashboard updates in real time with every transaction logged.
- Save: the app prompts the user to transfer a recommended tax provision into a dedicated Tax Savings Vault after each income entry.
- File: at filing season, the app auto-generates a pre-filled return from the full year of tracked data for GRA portal submission.
- Pay: the user pays via Mobile Money, bank card, or USSD - directly from the app or from the Vault.

Comparison: TaxPadi vs. Existing Options

Capability	TaxPadi	GRA App	Accountant
Tracks income and expenses in real time	Yes	No	Yes
Calculates tax liability automatically	Yes	No	Yes
Handles multiple tax types	Yes	Yes	Yes
AI-powered tax assistant (TaxBot)	Yes	No	No
Tax Savings Vault	Yes	No	No
Live penalty counter	Yes	No	No
Works fully offline	Yes	No	No
WhatsApp and USSD access	Yes	No	No
Affordable for informal operators	Yes	Yes	No

The Three AI Features

- Voice Input: powered by Google Speech-to-Text API. Users speak a transaction - "I received 500 cedis today" - and the app logs it automatically. Built for older, lower-literacy users who find typing cumbersome.
- Receipt Scanner: powered by Google Cloud Vision OCR API. Users photograph any receipt and the app extracts the amount, date, vendor, and category automatically. The same engine processes imported PDF bank statements.
- TaxBot: powered by the Anthropic Claude API constrained via system prompt to Ghana tax law and GRA regulations. Answers any tax question in plain language instantly. None of these require building or training a model - all three call existing external APIs from Spring Boot.

3. REQUIREMENTS

3.1 Functional Requirements

ID	Requirement	Priority
FR-01	User registers and completes tax profile: category, TIN/Ghana Card PIN, region, and industry	High
FR-02	App auto-assigns correct tax rate brackets, tax types, GRA deadlines, and deductible categories	High
FR-03	User logs income and expense entries manually with category, amount, date, and description	High
FR-04	Live tax liability meter updates in real time with every transaction logged	High
FR-05	Receipt scanner reads receipts via camera - extracting amount, date, vendor, and category using OCR	High
FR-06	Voice input converts spoken transaction descriptions into structured income or expense entries	High
FR-07	MoMo and bank statement import - user uploads file and app auto-categorizes all transactions	High
FR-08	Personal Income Tax calculated against Ghana's progressive brackets (0% to 35%) on cumulative income	High
FR-09	VAT threshold tracker alerts user approaching the GHS 200,000 registration threshold	High
FR-10	VAT Mode: calculates 21% output VAT on sales, input VAT on expenses, generates net monthly liability	High
FR-11	PAYE module: employee register, monthly PAYE calculation per employee, remittance deadline tracking	High
FR-12	Annual PAYE return auto-generated by March 31st from full year of employee salary and deduction records	High
FR-13	Smart tax deadline calendar with personalized reminders per tax type sent weeks in advance	High
FR-14	Tax Savings Vault: calculates recommended provision per income entry and prompts MoMo transfer	High
FR-15	Penalty calculator: live counter on missed deadlines, pre-deadline projection, and damage control guidance	High
FR-16	Auto-generated pre-filled tax return submitted via GRA portal redirect	High
FR-17	Tax payment via Mobile Money, bank card, or USSD within the app	High
FR-18	Digital compliance certificate generated after confirmed filing and payment	High
FR-19	TaxBot: AI assistant answering Ghana tax questions in plain conversational language	High
FR-20	Immutable audit trail: every transaction, filing, and payment timestamped and stored for 6 years	High
FR-21	No screenshot capture allowed on sensitive screens such as the tax dashboard and payment screens	High
FR-22	Data export in PDF, Excel, and JSON including a Verified Income Statement for loan applications	Medium
FR-23	Business Health Score: weekly 0-100 score based on income consistency, expense discipline, and compliance	Medium
FR-24	Withholding tax flagging: applicable transactions identified and remittance tracked	Medium

ID	Requirement	Priority
FR-25	Invoice Generator: user creates and sends professional invoices to clients with TIN, amount, and reference number. Marking an invoice as paid automatically logs it as income and updates the live tax liability meter	Medium
FR-26	Insurance Referrals: based on the user's verified income profile and business type, the app surfaces pre-qualified insurance products from partner providers. User is connected to the insurer directly within the app	Medium
FR-27	Lending Referrals: after 3 to 6 months of consistent income logging, the app surfaces pre-qualified loan offers from partner microfinance institutions based on the user's verified transaction history	Medium

3.2 Non-Functional Requirements

Category	Requirement
Performance	App loads within 2 seconds. Dashboard queries return within 1 second for up to 10,000 records.
Offline Support	Core features - logging, scanning, dashboard, penalty calculator - work without internet. Data syncs on connectivity restoration.
Scalability	Must support 50,000+ active users within 24 months without performance degradation.
Security	TLS 1.3 in transit, AES-256 at rest. JWT authentication, SMS OTP two-factor, biometric login. No screenshots on sensitive screens.
Data Privacy	Compliant with Ghana's Data Protection Act 2012 (Act 843). Registered with the Data Protection Commission before launch. Explicit user consent at onboarding.
Audit Integrity	All records immutable once created. Six-year retention aligned with GRA's audit window.
Reliability	99.5% uptime SLA. Automated daily backups with point-in-time recovery.
Usability	Onboarding completable in under 5 minutes. Voice and WhatsApp channels available as alternatives.
Localization	Ghana Cedis (GHS) throughout. Voice input supports Twi, Ga, Hausa, and Ewe.

4. ACTORS

Actor	Type	Description
Individual Taxpayer	Primary	Freelancer or self-employed person. Logs transactions, tracks liability, manages vault, files returns, and pays taxes.
Small Business Owner	Primary	Sole trader with employees. All individual functions plus PAYE management and VAT handling.
TaxPadi Admin	Internal	Platform administrator managing accounts, subscription tiers, and system health.
GRA	External	Ghana Revenue Authority. TaxPadi mirrors GRA filing formats and redirects to the GRA Taxpayers Portal for submission. E-Levy API integrated directly.
MTN MoMo / Telecel Cash	External	Mobile Money providers powering Tax Savings Vault transfers and tax payment processing.
Microfinance Partner	External	Licensed institution holding the Tax Savings Vault wallet and providing pre-qualified loan products to eligible users.

5. CORE ENTITIES

Entity	Key Attributes	Relationships
User	userId, fullName, tinOrGhanaCardPin, taxpayerCategory, subscriptionTier	Has many Transactions, TaxReturns, Payments, Employees, AuditLogs
Transaction	transactionId, userId, type, amount, category, entryMethod, date	Belongs to User; feeds TaxCalculation
TaxCalculation	calcId, userId, taxType, taxableIncome, taxLiability, calculatedAt	Belongs to User; derived from Transactions
TaxReturn	returnId, userId, taxType, period, taxLiability, status, submittedAt	Belongs to User; references TaxCalculation
Payment	paymentId, userId, returnId, amount, paymentMethod, transactionRef, paidAt	Belongs to User and TaxReturn
Employee	employeeId, userId, name, grossSalary, allowances, isActive	Belongs to User; has many PAYERecords
PAYERecord	payeId, employeeId, month, year, payeDeducted, remitted	Belongs to Employee
VATRecord	vatId, userId, month, year, outputVAT, inputVAT, netVATLiability	Belongs to User
SavingsVault	vaultId, userId, balance, linkedMoMoNumber, totalContributed	Belongs to User; linked to Payments
AuditLog	logId, userId, action, entityType, previousValue, timestamp	Belongs to User - immutable, append-only
Invoice	invoiceId, userId, clientName, description, amount, status, issuedAt, paidAt	Belongs to User; on payment status creates a linked Transaction

6. API DESIGN

TaxPadi exposes a RESTful API built with Spring Boot (Java), versioned under /api/v1 and secured with JWT Bearer tokens via Spring Security. Full documentation is maintained in Swagger / OpenAPI format.

Method	Endpoint	Description
POST	/auth/register	Register new user account with taxpayer profile
POST	/auth/login	Authenticate and receive JWT access and refresh tokens
GET / PUT	/users/me	Retrieve or update authenticated user profile and tax configuration
GET / POST	/transactions	List all transactions or log a new income/expense entry
POST	/transactions/import	Import and parse a MoMo or bank statement file (PDF or CSV)
POST	/transactions/scan	Submit receipt image for OCR extraction and auto-categorization
POST	/transactions/voice	Submit voice audio for speech-to-text transaction logging
GET	/tax/liability	Retrieve current live tax liability across all applicable tax types
GET	/tax/vat/status	Get VAT registration status, threshold progress, and monthly liability
GET / POST	/tax/paye/employees	List employees or add a new employee to the PAYE register
GET	/tax/returns/generate	Auto-generate a pre-filled tax return from tracked transaction data
POST	/tax/returns/submit	Mark a return as submitted and log GRA portal submission confirmation

Method	Endpoint	Description
POST	/payments/initiate	Initiate a tax payment via Mobile Money, bank card, or USSD
GET	/payments/certificate/{id}	Retrieve digital compliance certificate for a confirmed payment
POST	/vault/contribute	Initiate a MoMo transfer into the Tax Savings Vault
POST	/taxbot/ask	Submit a question to TaxBot and receive a Ghana tax answer
GET	/penalties/active	Retrieve active penalties with live running totals and guidance
GET	/reports/export	Export financial history as PDF, Excel, or JSON for a given period
GET / POST	/invoices	List all invoices or create a new invoice for a client
PUT	/invoices/{id}/paid	Mark invoice as paid and auto-log it as an income transaction
GET	/referrals/offers	Retrieve pre-qualified loan and insurance offers based on user income profile

7. LOW-LEVEL DESIGN

7.1 Tax Calculation Engine

All tax rates, brackets, thresholds, and deductible categories are hardcoded as application constants sourced from Ghana's Income Tax Act 2015 (Act 896), VAT Act 2013 (Act 870), and PwC Ghana's annually updated Tax Summary. The engine retrieves the user's taxpayer category and cumulative transaction totals, applies the correct rate structure, and recalculates the live liability on every transaction write. Constants are reviewed and updated annually after Ghana's national budget.

- Personal Income Tax: progressive brackets applied to cumulative net income after allowable deductions.
- VAT: output VAT at 21% effective rate on each sale; input VAT on eligible expenses; net = output minus input.
- PAYE: Ghana income tax brackets applied to each employee's monthly gross salary minus exempt allowances.
- Withholding Tax: applicable transaction categories flagged at entry with rate applied by transaction type.

7.2 Offline Sync Engine

The mobile app maintains a local copy of user data in SQLite. All core write operations execute against the local store first and are placed in a timestamped offline action queue. A Spring @Scheduled task processes the queue when connectivity is confirmed, replaying actions chronologically. Conflicts use a last-write-wins strategy with all overwritten versions archived in the AuditLog rather than deleted.

7.3 AI Integration Pattern

All three AI features follow the same Spring Boot pattern: receive client input, forward to the external API, parse the response, and return the result. Voice input calls Google Speech-to-Text; receipt scanning calls Google Cloud Vision OCR; TaxBot calls the Anthropic Claude API with a persistent system prompt constraining responses to Ghana tax law. No model training or hosting is required by TaxPadi.

7.4 Penalty Engine

A Spring @Scheduled daily job runs at midnight and queries all users with unresolved TaxReturn records past their due date. It calculates the current penalty using hardcoded GRA rates - GHS 200 base plus GHS 20 per day for personal income tax late filing; 10% of tax due plus 2% monthly interest for late payment - writes or updates a PenaltyRecord, and triggers an FCM push notification displaying the updated running total.

7.5 GRA Filing and Savings Vault Flow

On filing request, the backend aggregates the user's full period data, runs the final tax calculation, and generates a pre-filled return. The app opens the GRA Taxpayers Portal in an in-app browser for final submission. The E-Levy API is called directly for applicable transactions. For the Savings Vault: on each income entry, the engine calculates the recommended tax provision and surfaces a transfer prompt. On user approval, a payment request is sent to the MTN MoMo or Telecel Cash API to move the amount into the vault wallet held by the licensed microfinance partner.

7.6 Invoice Generator Flow

The user creates an invoice by entering the client name, service description, amount, and due date. The backend generates a unique invoice reference number and produces a formatted PDF the user shares via WhatsApp, email, or direct download. When the user marks an invoice as paid, the backend creates a corresponding Transaction record of type income, immediately triggering a recalculation of the live tax liability meter. The invoice is permanently archived in the audit trail.

7.7 Financial Referrals Engine

A Spring @Scheduled weekly job queries users who have logged at least 90 days of income history. For each eligible user it computes average monthly income, an income consistency score, and tax compliance rate. These values are matched against eligibility thresholds defined by each partner lender and insurer. Where a match exists, a pre-qualified offer is surfaced on the user dashboard. The user taps to review and is handed off to the partner provider via a deep link or in-app browser. TaxPadi earns a referral fee per successful handoff confirmed by the partner.

8. HIGH-LEVEL DESIGN

8.1 Architecture Overview

Component	Description
Mobile App	React Native with Expo (TypeScript). Handles UI, offline SQLite storage, push notifications, AI interfaces, and chart rendering. Syncs with backend on connectivity restoration.
API Layer	Spring Boot (Java). All business logic, JWT authentication, tax calculation engine, scheduled jobs, and AI API proxy calls.
Database	PostgreSQL - primary relational store for all user, transaction, tax, payment, PAYE, VAT, vault, audit, and penalty data.
Local Storage	SQLite on-device - mirrors core user data for full offline functionality and hosts the offline action queue.
Cache	Redis - caches tax liability calculations and report query results.
Background Jobs	Spring @Scheduled tasks for nightly penalty calculation, deadline reminder dispatch, and sync queue processing.
Push Notifications	Firebase Cloud Messaging (FCM) for deadline reminders, penalty alerts, and sync confirmations.
SMS	Arkesel (Ghana) for two-factor OTP and critical deadline SMS fallback.
Hosting	AWS EC2 (API), AWS RDS PostgreSQL, AWS ElastiCache (Redis). Containerized with Docker.

8.2 Technology Stack

Concern	Technology
Mobile Frontend	React Native + Expo (TypeScript)
Backend API	Spring Boot (Java)
Database	PostgreSQL
On-Device Storage	SQLite
Cache	Redis
Authentication	JWT via Spring Security + SMS OTP (2FA)
Voice AI	Google Speech-to-Text API
Receipt Scan AI	Google Cloud Vision OCR API

Concern	Technology
TaxBot AI	Anthropic Claude API
Push Notifications	Firebase Cloud Messaging (FCM)
SMS	Arkesel / Twilio
Payments	MTN MoMo API + Telecel Cash API
Hosting	AWS (EC2, RDS, ElastiCache)
Containerization	Docker

8.3 Business Model

Layer	Revenue Stream	Mechanism
1 - Freemium	Free core tier	Basic tracking and live liability meter free - drives user acquisition
2 - Subscription	Monthly or annual fee	VAT, PAYE, auto-filing, and advanced features behind paywall
3 - Transaction Fee	Payment processing fee	Small flat fee on every tax payment processed through the app
4 - Financial Services	Lending and insurance referral fees	Verified income data used to surface pre-qualified loan and insurance products from partner providers. TaxPadi earns a referral fee on every successful connection
5 - Enterprise	B2B white label	Enterprise version sold to medium and large companies on annual contracts
6 - Data Intelligence	Anonymized SME data	Aggregated income and spending patterns sold to banks, insurers, and policy bodies at scale

9. GROWTH STRATEGY

Phase	Timeline	Strategy
Phase 1 - Adoption	Months 1-12	Free tier with core tracking and live tax meter. Agent network at markets and GRA service centers. WhatsApp bot and USSD activated. GRA and MTN MoMo partnership discussions initiated.
Phase 2 - Monetization	Months 6-18	Paid subscription launched: VAT, PAYE, auto-filing, and Tax Savings Vault unlocked. Microfinance lending partnerships activated.
Phase 3 - Integration	Months 12-24	Direct MTN MoMo API for automated vault provisioning. GRA API partnership formalized. Corporate tier for registered companies launched.
Phase 4 - Data Revenue	Month 24+	Anonymized SME financial data product launched. B2B enterprise white label sales commenced.

10. FUTURE ENHANCEMENTS

The following capabilities are excluded from Version 1.0 and represent the confirmed product roadmap beyond launch:

- Direct GRA API Integration: Version 1.0 redirects users to the GRA Taxpayers Portal for return submission. A formal API partnership with GRA will enable direct programmatic submission from within TaxPadi. The existing E-Levy API serves as the initial proof of concept and foundation for that conversation.
- Accountant Marketplace: a verified in-app marketplace of ICAG-registered tax consultants. Users hire accountants directly, grant granular and revocable data access permissions, and collaborate through a

shared workspace with messaging, task management, and document sharing. TaxPadi earns a platform commission on every engagement.

- Registered Company and Corporate Tax Tier: a Premium Business Tier for formally registered companies handling corporate income tax at 25%, quarterly payment scheduling, multi-user access for finance teams, and payroll management at scale.
- Direct MoMo API Transaction Sync: Version 1.0 requires manual MoMo statement import. Phase 2 introduces direct MTN MoMo API integration pulling transactions automatically in real time with user authorization, eliminating manual imports entirely.
- Full Local Language Interface: a complete Twi interface making the app fully accessible to non-English users, with further languages added based on user distribution data.
- Cash Flow Forecasting: a 30-day forward projection engine built from 90 days of historical income and expense data, flagging weeks where projected expenses exceed projected income as financial risk periods.
- Anonymized SME Financial Data Product: at sufficient scale, aggregated and anonymized income, spending, and compliance patterns sold as market intelligence to banks, insurance companies, and policy research bodies. No real-time data source for Ghana's informal economy currently exists - TaxPadi at scale becomes the only platform with it.

CONCLUSION

TaxPadi addresses a real, widespread, and consequential gap in Ghana's financial ecosystem. Millions of self-employed individuals, traders, and small business owners contribute meaningfully to the national economy yet lack the tools to meet their tax obligations accurately, proactively, and without unnecessary cost. The GRA has made commendable progress in digitizing tax filing and payment - but the step before filing, helping users understand and prepare their financial data throughout the year, remains entirely unaddressed.

TaxPadi fills that gap with a product that is intelligent enough to automate the hard parts, simple enough for a market trader to use, and robust enough to withstand a GRA audit. It manages five distinct tax types, integrates three AI capabilities, operates fully offline, and is built on a multi-layered revenue model with clear phased growth logic. Every design decision traces back to one user: the Ghanaian who is willing to comply but has never had the right tool to do so.

We are confident that TaxPadi represents a viable, impactful, and technically sound software solution. We respectfully submit this proposal for the consideration of the Codequest panel and welcome any questions or feedback.

TaxPadi | Because GRA doesn't forget.

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